Issa Aziz KARAMBIRI

Engineering student in applied mathematics and computer science, ENSIIE (France)
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Profile

Second-year engineering student at ENSIIE (Master 1), studying applied mathematics and computer science with a focus on data, AI and finance. My projects have mostly shown me how much I still have to learn, and that is exactly what motivates me. I ask questions, I try, I sometimes get it wrong, and I learn fast. I enjoy concrete problems, demanding teams and taking on responsibility.

Experience

Data science intern, OptiTrade – ENTSI, Oujda (Morocco) Mar – May 2025

- Investment support web app: 300,000 rows of stock market data at the start, 246,437 after cleaning.
- Compared four models (linear regression, Random Forest, XGBoost, LSTM) with MSE, RMSE and R^2 to recommend buying, selling or holding a stock.
- Python, Django REST, React, PostgreSQL, Scikit-learn, TensorFlow.

Web developer intern, GESCOM – ENTSI, Oujda (Morocco) Jun – Jul 2024

- Sales management app (customers, products, sales, purchases, user roles) in PHP, MySQL, Bootstrap and jQuery.
- Optimised the slowest SQL queries.

Projects

Predictive maintenance of aircraft engines – machine learning, NASA data GitHub

- Predicts the number of flights left before failure (XGBoost, typical error of 11 flights), with an inspection threshold chosen to minimise total cost.
- 25 out of 25 imminent failures detected, with only 5 false alarms on 75 healthy engines.

VIGIE, anti-scam assistant for seniors – Project Challenge, 4 days

- Team of 5 students from 3 schools (ENSIIE, Télécom SudParis, IMT-BS). My role: technical and prototyping lead.
- Fraudulent link detection, warning before the click, and screens readable by a 74-year-old.

Bank fraud detection – highly imbalanced data GitHub

- Fraud makes up 0.17 % of transactions. XGBoost tuned in terms of euros saved (PR-AUC 0.88).
- Fraud losses cut by 77 %, with 87 % of alerts correct.

Beach litter detection – deep learning Jan – Mar 2025

- Hand-labelled images (6 classes), data augmentation, YOLOv8 (PyTorch) training on GPU and an OpenCV interface.
- Precision 89.4 %, recall 89.5 %, F1-score 89.5 %.

School projects: Black-Scholes model in C++ (European option pricing), OCaml virtual machine in C (bytecode, stack, memory), mini relational DBMS in OCaml.
All my projects: issakarambiri.com

Education

École Nationale Supérieure d'Informatique pour l'Industrie et l'Entreprise (ENSIIE) since Sept. 2025
Engineering programme, 2nd year (Master 1), Évry-Courcouronnes, France. Probability, stochastic processes, optimisation, operations research, machine learning, deep learning, SQL and NoSQL databases.

DUT, highest honours – École Supérieure de Technologie d'Oujda (Morocco) 2023 – 2025

Skills and interests

Languages: Python, SQL, C++, C, R, OCaml, Java, PHP, HTML, CSS, JavaScript
Data and AI: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, PyTorch, YOLOv8, OpenCV, SHAP
Tools: PostgreSQL, Django, Git, VS Code, Jupyter, Google Colab, LaTeX, Power BI, Excel
Spoken languages: French (native), English (B2)
Interests: reading, music, cooking (always looking for new recipes)

Volunteering and awards

Ambassador, YITU Association – tutoring and CV workshops for students since Aug 2026
Academic officer, AEBM – coordinated the association's academic activities 2024 – 2025
Public speaking contests – winner (CETASSO, Oujda, 2024), finalist (AEBM, 2025) 2024 – 2025
Excellence awards and certificates – yearly average above 16/20 (CETASSO, AEBM, UEEMO) 2023 – 2024